

[Home](#) > Floods in India

# Floods in India, Causes, Impacts, Mitigation, Measures

Floods are natural disasters that occur when a large amount of water submerges normally dry ground. India is highly vulnerable to floods in which Assam, Bihar, Odisha, Uttar Pradesh, and West Bengal are largely affected.

By **Vajiram Editor** - Jul 27, 2026, 12:58 IST

English ▼



Share

Select as

[Table of Contents](#)

Floods are the most common form of natural disaster, occurring when a large amount of water submerges normally dry ground. Floods are frequently produced by excessive rainfall, quick snowmelt, or storm surge from a tropical cyclone or tsunami in coastal locations. India is highly vulnerable to floods in which Assam, Bihar, Odisha, Uttar Pradesh, and West Bengal are largely affected. Flood disasters affect people and the impact of floods cannot be completely prevented, while measures can be taken to mitigate the impact and ensure that people are prepared to deal with the consequences of the flood hazards.

## Flood-prone Areas of India

[Enquire Now](#)

India is the worst-affected country by floods after Bangladesh, accounting for one-fifth of all flood-related deaths worldwide. According to the Ministry of Jal Shakti, the recent estimate on flood-prone areas in the country is 49.15 Mha. The annual average cropped area affected by flood is approximately 3.7 million hectares.

- **Indo-Gangetic- Brahmaputra plains:** The most flood-prone areas in India are the Brahmaputra and the Ganga River basins in the Indo-Gangetic- Brahmaputra plains in North and Northeast India, which carry 60 percent of the nation's total river flow.
- **Affected states:** Nearly 75% of total Indian rainfall occurs during a brief monsoon season (June-September). As a result, the rivers witness a heavy discharge during these months, leading to widespread floods in Uttar Pradesh, Bihar, West Bengal, and Assam.
  - Assam Valley is highly flood-prone, primarily due to heavy rains and silting of tributaries like Dihang and Lohit.
  - In Kashmir Valley, the Jhelum is unable to carry the river discharge.
  - In Punjab and Haryana, the problem is primarily due to inadequate drainage.
  - The Central Indian and peninsular river basins, covering Madhya Pradesh, Orissa, Andhra Pradesh, Maharashtra, and Gujarat, contain the Tapi, Narmada, and Chambal rivers, which often experience excess rainfall.
  - In Andhra Pradesh, the Kolleru lake submerges vast areas along its fringes and cyclonic storms bring heavy flooding in the coastal areas.
- **Flash flood-prone areas:** In recent decades, states such as Rajasthan, Gujarat, Haryana, and Punjab have been inundated by flash floods.

## Causes of Floods in India

India is prone to floods due to its geography, heavy rainfall during the monsoon and global warming leading to more frequent and intense rainfall events, etc. Floods are caused by both natural as well and anthropogenic factors.

### Natural Causes

- **Heavy rainfall:** Heavy rain in the catchment area of a river causes water to overflow its banks, which results in the flooding of nearby areas.
  - During the monsoon season, there are many events of flood in northern and northeastern India and the Gangetic plains due to heavy rainfalls.
- **Cloud Burst:** Cloud Burst causes intense precipitation in a very short duration which can cause a flood.
  - These natural incidents are observed in the Himalayan regions.



**Enquire Now**

- **Sediment deposition:** Large silts and sediments get deposited in the river bed, which reduces the carrying capacity of rivers.
- **River channel changes:** Natural processes such as erosion and meandering can alter river channels over time. These changes can affect the capacity of rivers to carry water, leading to increased flood risk.
  - **River Kosi** in Bihar is known as the “**sorrow of Bihar**” due to frequent floods caused by its nature of changing course.
- **Cyclones:** India's coastal regions, especially along the Bay of Bengal and Arabian Sea, are susceptible to tropical cyclones, which can cause significant flooding due to heavy rainfall and storm surges.
  - In December 2023, Cyclone Michaung made landfall in Andhra Pradesh resulting in storm-induced floods, leading to casualties and damage to properties.
- **Himalayan glacier melt:** The melting ice can result in the formation of glacial lakes, which pose a risk of **glacial lake outburst floods (GLOFs)**. Example: **Uttarakhand Glacial burst**.
- **Climate Change:** Climate change has aggravated the situation of flood risks.
  - In mountainous regions, three effects of climate change are there -more intense precipitation due to a warmer atmosphere, shifting snow and rain patterns, and the effects of wildfires on the landscape.
  - Climate change is also causing intense rain and the occurrence of tropical rainfall in the Indian Ocean region. Thus, it has also caused an increased risk of floods in the Western Ghat region.

## Anthropogenic Causes

- **Deforestation:** Vegetation plays a crucial role in absorbing rainwater and reducing surface runoff. Deforestation disrupts natural rainwater absorption and reduces surface runoff, increasing runoff volumes and river flooding risk.
  - According to a study, with every **10 percent increase** in trees being cut down, the flood risk increases by up to **28 percent**.
- **Poor drainage:** Urban growth converts natural land into impermeable surfaces. This reduces land's ability to absorb water and leads to increasing surface runoff, and insufficient or poorly maintained drainage systems increase flood risk.
  - For instance, **recent floods in Bangalore** are more attributed to the reasons for poor drainage and urbanisation.
- **Dam and embankment failure:** Dam collapse and river embankment failures due to poor infrastructure can also cause floods.
  - In case of heavy rainfall and limitation of water storage capacity of dams, the dam gates are opened which causes severe floods in the nearby areas.
- **Encroachment on floodplains:** Construction and settlement in floodplains obstructs the natural flow of water during floods.



**Enquire Now**

- Encroachment **reduces the capacity of floodplains** to store excess water, causing it to overflow into surrounding areas.
- **Encroachment of water bodies:** Encroachment of lakes, ponds and other water bodies aggravates the risk of floods because they cannot contain the extra water when the situation of flooding arrives.

&lt;

&gt;

## Impacts of Floods in India

Floods are generally destructive to humans and the natural environment, but they also contribute positively in some cases.

### Negative Impact

- **Immediate impacts:** It causes loss of life and property, crop destruction, livestock loss, infrastructure malfunction, disruption in movement and deterioration of health conditions due to waterborne diseases.
  - According to Central Water Commission (CWC) data, in the last 65 years, floods killed **109,412 people**, and 258 million hectares of crops were damaged.
- **Economic loss:** Floods cause damage to infrastructure, leading to livelihood loss and disruption in industries and businesses, causing production delays, supply chain disruptions, and financial losses.
  - According to SBI research, floods in North India caused an estimated economic loss worth ₹10000-15000 crore in 2023.
- **Displacement:** Floods result in a humanitarian crisis and displacement of the people.
  - Flooding in Jammu and Kashmir in 2015, Uttarakhand in 2013, and Assam in 2012 displaced **millions of people**.
- **Health risks and waterborne diseases:** Floods contaminate water sources, leading to an increased risk of waterborne diseases such as **cholera, typhoid, and diarrhoea**.
  - The **World Health Organization** estimates that floods in India result in the loss of millions of productive days and contribute to a substantial **disease burden**.
- **Environmental impacts:** Floods cause soil erosion, sedimentation, and de
  - Sediment-laden floodwaters can damage agricultural land, affecting

**Enquire Now**

biodiversity.

## Positive Impact

- **Recharging water sources:** Inundation of the flood plains helps recharge the groundwater, which is an important source of drinking water and is essential for agriculture.
- **Agriculture:** Floodwaters carry nutrients and sediments, which are deposited on flood plains, enriching the soil. Rice paddies are flooded deliberately to take advantage of this natural fertilisation process.
- **Rejuvenation of the river ecosystem:** The river ecosystem is crucial for the biodiversity of fish, wildlife, and waterfowl, and its seasonal variability and varied sediment and flow regimes help maintain this balance.
- **Fishery:** Floods trigger aquatic species spawning and migration, providing abundant fish supply and alternative income sources at the household level.



Don't miss Vajiram & Ravi updates:

Add Vajiram & Ravi as a preferred source on Google

Add →

## India's Preparedness for Floods

The Indian government has implemented many initiatives and policies to lessen the impact of floods through various state and central disaster management support organisations.

- **Central Water Commission (CWC):** It was set up in 1945 and focuses on flood control, water resource conservation, irrigation, hydropower generation, flood management, and river conservation in the country..
- **National Disaster Management Authority (NDMA):** NDMA has prepared these **Guidelines for Flood Management**, to assist the ministries and departments of the GOI, the state governments, and other agencies in preparing Flood Management plans (FMPs).
- **The National Water Policy (2012):** It emphasises flood control through structural measures, integrated reservoir operation, natural drainage system rehabilitation, farming systems, and non-agricultural development, aiming for long-term solutions to devastating floods.
- **Flood Forecasting Network:** The CWC has implemented a flood forecasting system with 175 stations across major interstate rivers, enabling real-time data collection, automatic transmission, flood forecast formulation, and efficient information dissemination.
- **Flood Management and Border Areas Programme (FMBAP):** It is being implemented throughout the country for effective flood management, erosion control, and anti-sea erosion to maintain peace along the border.



Enquire Now



- **Structural Measures:** Since 1954, the government has constructed 33,928 km of new embankments and 38,809 km of drainage channels, completed 2,450 town protection works, and raised 4,721 villages above flood levels.
- **Post-disaster response and recovery:** India has set mechanisms for the post-disaster response, such as search and rescue operations, relief and rehabilitation camps, NDRF forces, and the use of **earth observation satellites**.
- **Cooperation with the neighbouring countries:** India has cooperation with neighbours like Nepal, China, and Bhutan for the exchange of hydro-meteorological data for early preparedness of flood management.

## Challenges with Flood Management in India

Despite significant investments and ongoing flood management measures in India, socioeconomic damages and mortality tolls remain high. The following are the challenges of flood management in India:

- **Complex hilly regions:** Flood risk management in hilly regions is still in the infancy stage, particularly due to complex and tough terrain with limited accessibility and a low level of monitoring.
- **Technological problems:** Innovative technologies are emerging to help manage flood risk, but these are not always straightforward to implement and technology alone will not address all our challenges.
- **Lack of policy implementation:** Though non-structural measures (such as regulations, land use planning, flood zoning, and flood forecasting) find their mentions in flood-related policies and are practised in a few regions of the country, they are yet to be enacted extensively by the States.
- **Addition of new flood-prone areas:** Flood-prone regions are growing rapidly due to climate and human factors, posing challenges for organizations to implement effective plans and policies to minimize human and economic losses during and after disasters.
- **Policy failure:** Over the years, the National Flood Commission recommendations have failed to achieve much success in mitigating floods, and the data show that flood damage and fatalities have only increased for years.

## Suggested Measures

A NITI Aayog report on flood management under Rajiv Kumar has suggested effective and long-lasting strategies for flood control and management that involve structural and non-structural measures along with the use of modern technologies.

- **Non-structural measures:**
  - **Implementation of floodplain zoning:** It has been suggested to **main** zoning approach as mandated by the Ministry of Jal Shakti.



**Enquire Now**

- **Scientific research:** It has called for modernisation in the collection of **hydrometeorological data**, flood forecast formulation, forecast dissemination and focus on scientific research in the development of a Model-based system to forecast flash floods.
- **Use of space technology:** Emphasis is to be laid on the application of **space technology** and the use of **state-of-the-art** technology in **spatial flood early warning** and near real-time monitoring and mapping of floods.
- **Structural measures:**
  - **Storage reservoirs:** The construction of large storage reservoirs that moderate flood peaks is the need of the hour.
  - **Embankments:** The raising and strengthening of existing **embankments** and creating new embankments, channelisation, drainage & channel improvement, and anti-erosion works have been suggested.
  - **Interlinking of rivers:** The projects for **interlinking of rivers** for diversion of flood water to water-scarce regions should be completed.
  - **Emergency Action Plans** for all big dams should be prepared and implemented when needed.

### Latest UPSC Exam 2026 Updates

*Last updated on August, 2026*

- **UPSC Mains 2026** will be conducted on 21st, 22nd, 23rd, 29th and 30th August 2026.
- Check out the latest **UPSC Syllabus 2026** here.
- **UPSC Mains Admit Card 2026** is expected to be released soon at [upsc.gov.in](https://upsc.gov.in) or [upsconline.nic.in](https://upsconline.nic.in)
- Enroll in Vajiram & Ravi's **UPSC Mains Test Series 2027** for structured answer writing practice, expert evaluation, and exam-oriented feedback.
- Join Vajiram & Ravi's **UPSC Mentorship Program 2027** for personalized guidance, strategy planning, and one-to-one support from experienced mentors.
- Go through the **UPSC Mains Previous Year Papers** to enhance your preparation.
- Download **UPSC Mains Essay Paper 2025**, **UPSC Mains GS Paper-I 2025**, **UPSC Mains GS Paper-II 2025**, **UPSC Mains GS Paper-III 2025**, **UPSC Mains GS Paper-IV 2025**, **UPSC Mains English (Compulsory) Paper 2025**, **UPSC Mains Hindi (Qualifying) Paper 2025**



**Enquire Now**

→ UPSC has released [UPSC Toppers List 2025](#) with the Civil Services final result on its official website.

→ [UPSC Calendar 2027](#) has been released.

→ Also check [Best UPSC Coaching in India](#)

# Floods in India FAQs

Q1. What are the causes of floods?



Q2. What are the impacts of flood?



Q3. How are floods predicted?



Q4. Which are the most flood-prone areas of India?



Q5. Why India is more prone to floods?



Tags: [floods in india](#) [quest](#)



Download this Article as PDF

Previous Article

[National Security Act 1980, Objectives, Provisions, Criticism](#)

Next Article

[Landslides, Types, Causes, Impacts, Mitigation, Prone Areas](#)

## Vajiram Editor

At Vajiram & Ravi, our team includes subject experts who have appeared for the UPSC Mains and the Interview stage. With their deep understanding of the exam, they create content that is clear, to the point, reliable, and helpful for aspirants. Their aim is to make even difficult topics easy to understand and directly useful for —whether it's for Current Affairs, General Studies, or Optional subjects. Ever designed to save your time and boost your performance.



Enquire Now



Related Posts

---

**National Pipeline**

Nilesh Dhamane | August 12, 2026

**Infrastructure****Galwan Valley Conflict**

Vajiram Editor | August 12, 2026

**Third Battle of Panipat 1761**

Vajiram Editor | August 12, 2026

Our Latest UPSC Courses

---

**SC Mentorship Program****₹1000****Enroll Now****UPSC Sureshot Mains Test Series****₹27000****Enroll Now****Prelims Powerup Test Series****₹14000****Enroll Now**



**VAJIRAM & RAVI**  
Institute for IAS Examination

# SURE SHOT

## MAINS TEST SERIES 2027

(Rankers Led Test Series)

Your preparation today shapes the  
India of tomorrow.

Tested Before Trusted

Starts  
**12<sup>th</sup>**  
Aug, 2026



Mode:  
Offline | Online



PRACTICE WITH FOCUS

IMPROVE EVERYDAY

SERVE THE NATION



Enquire Now



**VAJIRAM & RAVI**  
Institute for IAS Examination

# GENERAL STUDIES

## PRELIMS CUM MAINS PROGRAMME

**FOR UPSC 2027**

Understanding **India** In Every Dimension.

**STARTS**  
**25<sup>th</sup>**  
**August 2026**

**MODE**  
**Classroom & LIVE-Online**



**Enquire Now**

### Recent Posts

---

National Infrastructure Pipeline, Significance, Challenges

UPSC Mains Admit Card 2026, Download Mains Hall Ticket PDF

Galwan Valley Conflict 2020, Background, Causes, UPSC Notes

UPSC Exam Pattern 2026, Prelims, Mains & Interview

Champaran Satyagraha 1917, Leaders, Significance, UPSC Notes



**Enquire Now**



**Enquire Now**

## UPSC Exam 2026

---

UPSC Notification 2026

UPSC 2027

UPSC Calendar 2027

UPSC Prelims 2027

UPSC Mains 2026

UPSC Apply Online

UPSC Age Limit

UPSC Syllabus

UPSC Mains Syllabus

UPSC Books

NCERT Books for UPSC



**Enquire Now**



UPSC Optional Subjects List

IAS Salary

UPSC Previous Year Question Papers

UPSC Admit Card

UPSC Prelims Result

UPSC Mains Result

UPSC Final Result

UPSC Cut Off

IAS vs IPS

### UPSC Courses

---

UPSC GS Course 2027

UPSC GS Online Course 2027

UPSC 2 Year Course

UPSC 2 Year Online Course

UPSC Mentorship Program 2027

UPSC Mentorship Program Hindi Medium

UPSC Mains Test Series 2027

UPSC Mains Test Series Hindi Medium

UPSC Prelims Test Series 2027

UPSC CSAT Course

UPSC Online Coaching

UPSC Study Material

UPSC Optional Test Series

UPSC Interview Guidance Program IGP

Current Affairs Course

### UPSC Notes

---

Ancient History Notes for UPSC

Medieval History Notes for UPSC

Modern History Notes for UPSC

World History Notes for UPSC

Post-Independence India Notes for UPSC



**Enquire Now**

[Art & Culture Notes for UPSC](#)

[Geography Notes for UPSC](#)

[Indian Society Notes for UPSC](#)

[Governance Notes for UPSC](#)

[Polity Notes for UPSC](#)

[Social Justice Notes for UPSC](#)

[International Relations & Institutions Notes for UPSC](#)

[Economy Notes for UPSC](#)

[Agriculture Notes for UPSC](#)

[Science and Technology Notes for UPSC](#)

[Environment & Ecology Notes for UPSC](#)

[Disaster Management Notes for UPSC](#)

[Internal Security Notes for UPSC](#)

[Ethics Notes for UPSC](#)

## About Vajiram & Ravi

[About Us](#)

[Disclaimer](#)

[Contact Us](#)

[Terms and Conditions](#)

[Privacy Policy](#)

[Careers](#)

[FAQs](#)

## About UPSC Exam

[UPSC Notification 2026](#)

[UPSC 2027](#)

[UPSC Syllabus](#)

[UPSC Syllabus in Hindi](#)

[UPSC Exam Pattern](#)

[IAS Salary](#)

[UPSC Age Limit](#)

[UPSC Calendar 2027](#)

[UPSC Interview](#)

[UPSC Optional Subjects](#)



**Enquire Now**

[UPSC Selection Process](#)

[UPSC CSAT Syllabus](#)

[UPSC Books](#)

[UPSC Previous Year Papers](#)

[UPSC Final Result 2025](#)

[UPSC Cut Off](#)

[UPSC Current Affairs](#)

[UPSC Answer Key](#)

[UPSC Admit Card](#)

## UPSC Courses

[UPSC GS Course 2027](#)

[UPSC GS Online Course 2027](#)

[UPSC 2 Year Course](#)

[UPSC 2 Year Online Course](#)

[UPSC Mentorship Program 2027](#)

[UPSC Mentorship Program Hindi Medium](#)

[UPSC Mains Test Series 2027](#)

[UPSC Mains Test Series Hindi Medium](#)

[UPSC Prelims Test Series 2027](#)

[UPSC CSAT Course](#)

[UPSC Online Coaching](#)

[UPSC Study Material](#)

[UPSC Optional Test Series](#)

[UPSC Interview Guidance Program IGP](#)

[Current Affairs Course](#)

## Install App

## Connect on Social Media

© 2026 Vajiram & Ravi • All rights reserved.



**Enquire Now**